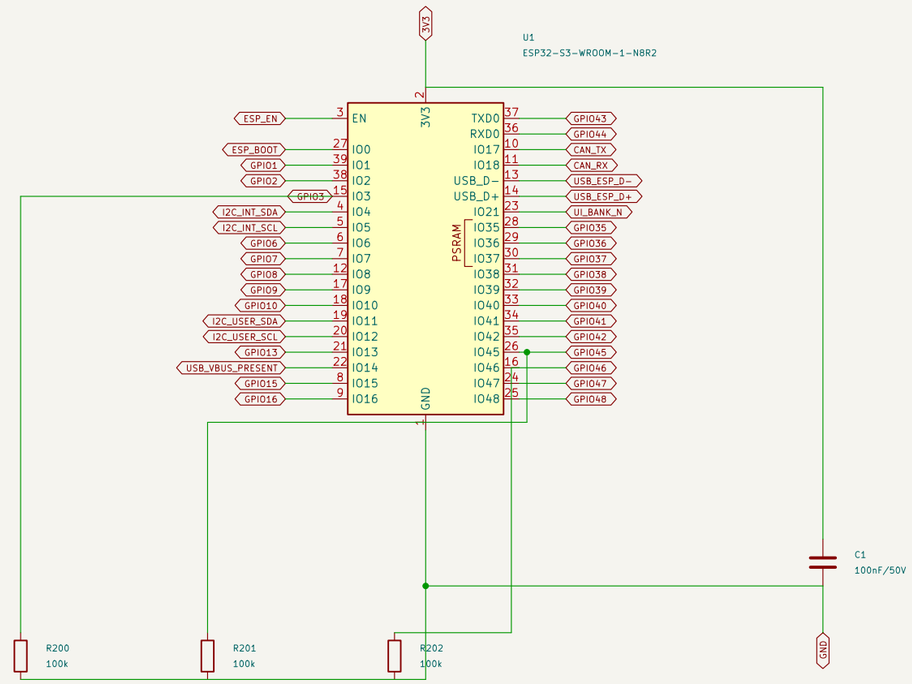
SPAGHETTI LAB | 16 RIGHE | ESP32-S3 N8R2

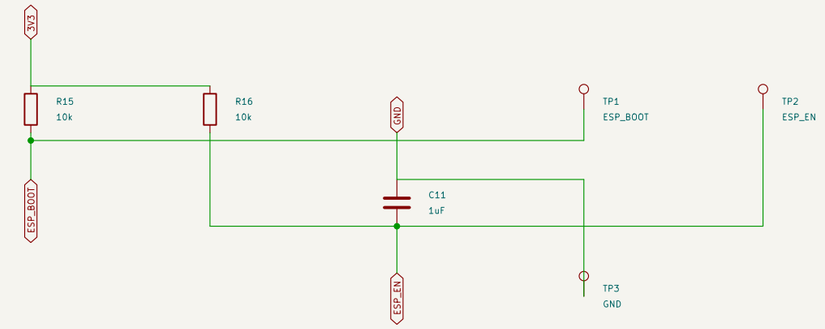
ESP32 / FLEXY PIN

GPIO1..10: ADC1. GPIO35..42: digitali. USB e CAN dedicati.



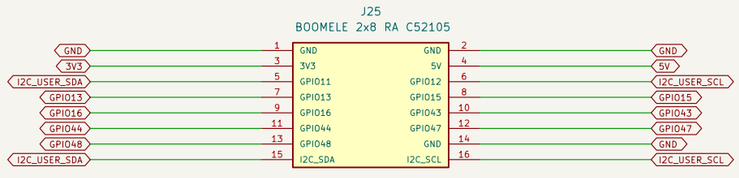
BOOT, EN E PIAZZOLE DI SERVIZIO

ESP_BOOT=strap GPIO0: ESP_EN=reset. Pull-up mantenuti.



ESPANSIONE SUD / BOOMELE C52105 2x8

GPIO diretti 3.3V. SDA/SCL condivisi con Qwiic ed expander.



Pin1 quadrato = GND. Passo e distanza tra file: 2.54mm.

5V solo alimentazione; GPIO non tolleranti 5V. Nessuna commutazione delle righe.

LEGENDA COLLEGAMENTI GLOBALI
3V3: rail regolata; 5V: rail ingresso; GND: riferimento comune.
I2C_INT: GPIO4/5 solo expander; I2C_USER: GPIO11/12 connettori; CAN_TX/RX: transceiver.
USB_ESP_D+/D-: coppia USB verso ESP; USB_VBUS_PRESENT: rilevamento5VUSB.
RGB_3V3_DATA: uscita ESP per LED; EXP_RESET_N: reset comune expander.
ROW_Lxx = banco A; ROW_Rxx = banco B. Suffissi spiegati nel foglio05.
Tutti i nomi sono globali; i fili mostrano la topologia. Punto = connessione, incrocio senza punto = nessun collegamento.

02 – Alimentazione e USB

File: power-usb.kicad_sch

03 – CAN e terminazione automatica

File: can.kicad_sch

04 – Controllo righe, reset sicuro e Qwiic

File: control.kicad_sch

05 – Sedici righe programmabili

File: ports16.kicad_sch

06 – Indicatori di configurazione retroilluminati

File: rgb-ring.kicad_sch

07 – Mezzanina passiva

File: mezzanine.kicad_sch

08 – Selezione, comandi e logo

File: ui-indicators.kicad_sch

09 – Row touch,16keys

File: touch16.kicad_sch

SELECT=GPIO21 / UI_BANK_N: OK=GPIO13. Both inputs active low with internal pull-ups.
Four persistent slots: two-button UI contract in SELECT-OK-UI.md. Global icons are LED indicators, not touch keys.

Sheet: /
File: backbone.kicad_sch

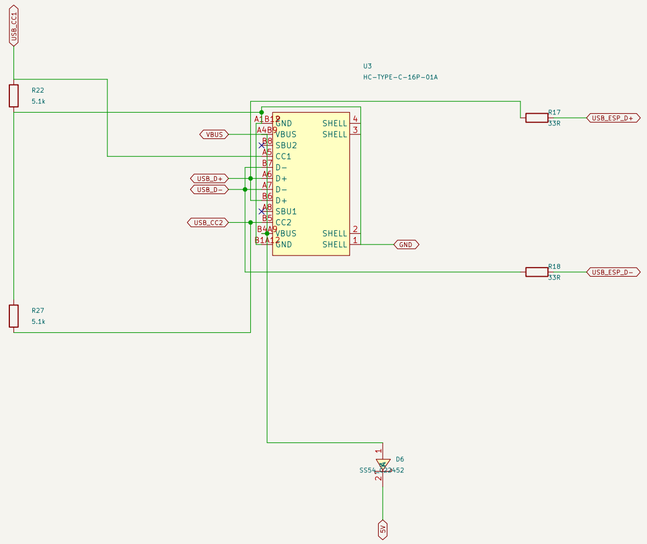
Title: 01 – ESP32 e boot

Size: User | Date: 2026-11-09 | **Rev: 16R-SELECTOK4**
KiCad E.D.A. 10.0.3 | Id: 1/9

USB-C E ALIMENTAZIONE 5V / 3V3

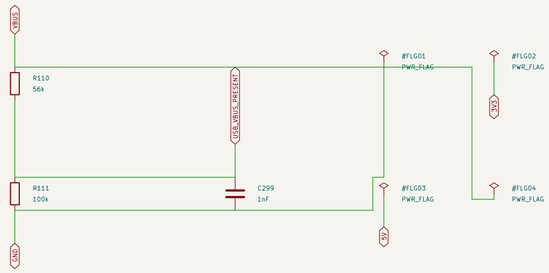
USB-C, CC E DATI

CC1/CC2: resistenze dispositivo USB; D6: alimentazione da VBUS.



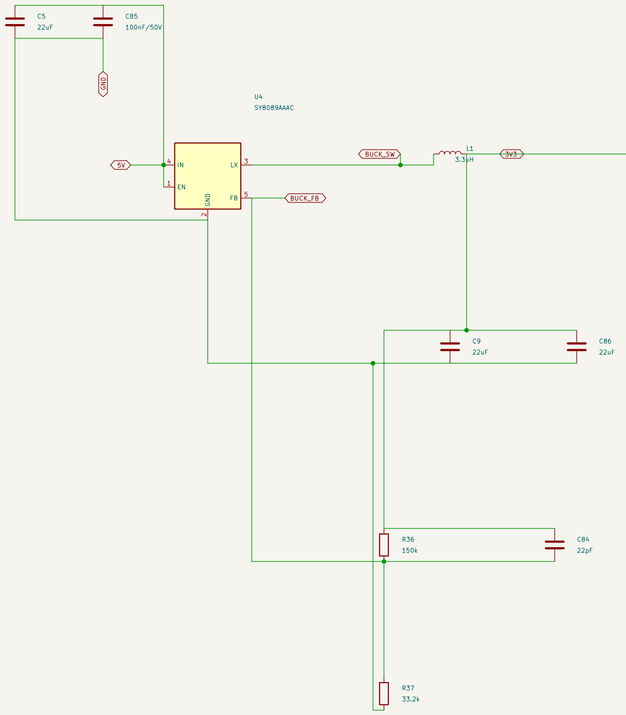
RILEVAMENTO VBUS E FLAG ALIMENTAZIONI

USB_VBUS_PRESENT: partitore56k/100k verso GPIO14, filtro1nF.



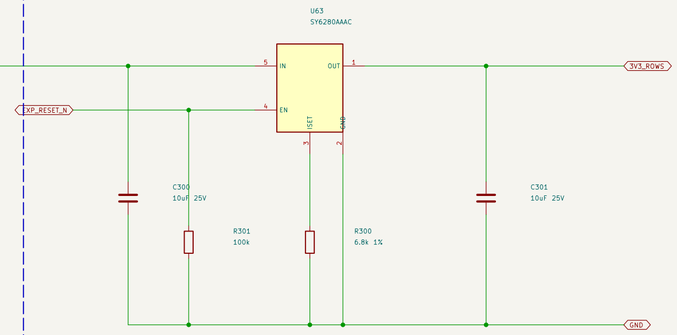
REGOLATORE 5V -> 3V3

BUCK_SW: nodo commutato; BUCK_FB: retroazione del regolatore.
Le etichette 5V/3V3/GND connettono gli altri fogli.



PROTEZIONE COMUNE ALIMENTAZIONE RIGHE

3V3_ROWS alimenta solo i rami VCC delle16righe.
ESP32, CAN, Qwiic e connettori diretti restano su3V3.



R300: ILIM=6800/R; 6.8k -> 1A nominale condiviso.
Datasheet: 0.75..1.25A a5V/25C; 500mA a 3.3V da validare.
Corto persistente: limite corrente, arresto termico e riprova.
Non protegge GPIO, 5V esterni o le rail dirette dei connettori.
Il chip non fornisce un segnale FAULT. EN non deve fluttare.

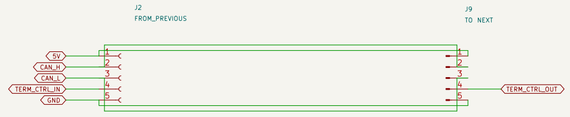
Avvio: EXP_RESET_N basso -> U63 spento.
Senza scarica attiva (versioneA): i condensatori possono restare carichi. Righe Hi-Z garantite solo dalle sequenze e dai pull del circuiti esistenti, da provare sul prototipo.
GND rimane comune. PCB ancora da sbrogliare.

Sheet: /02 – Alimentazione e USB/
File: power-usb.kicad_sch
Title: 02 – Alimentazione e USB
Size: A2 | Date: 2026-11-07 | Rev: 16R-READABLE
KiCad E.D.A. 10.0.3 | Id: 2/9

BUS CAN E TERMINAZIONE AUTOMATICA

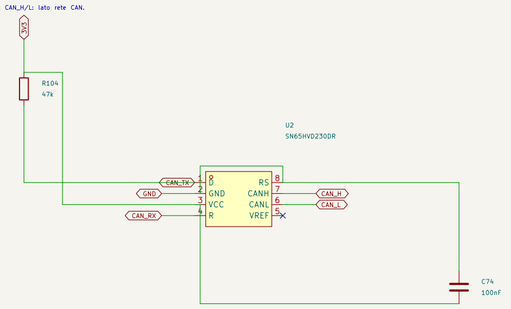
CONNETTORI MAGNETICI PASSANTI

CAN_H/L: bus differenziale. TERM_CTRL_IN/OUT: rilevamento dei vicini.



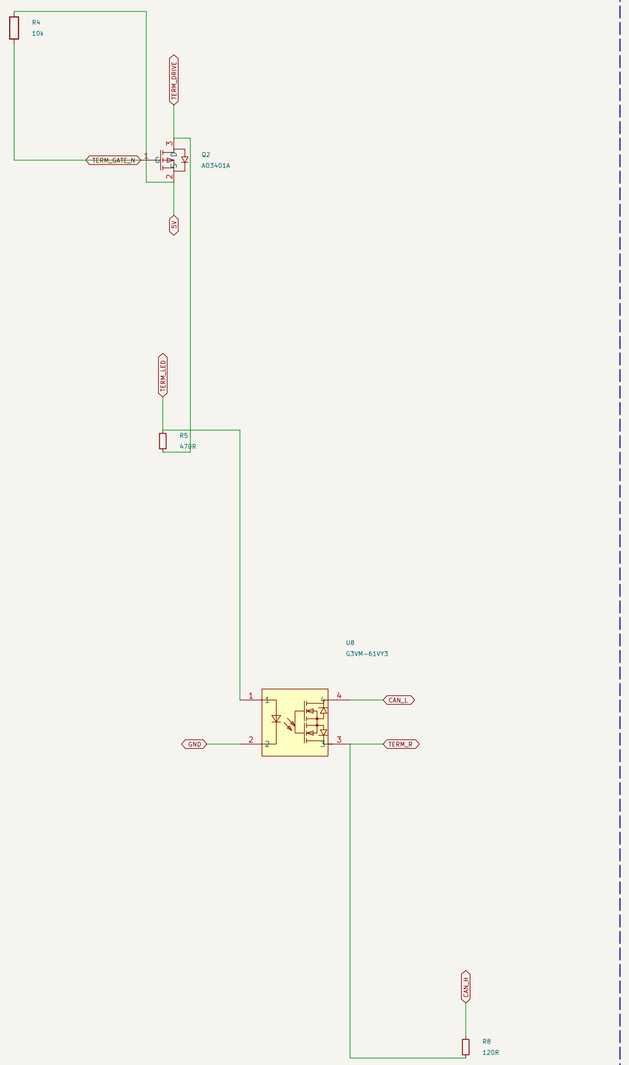
TRANSCEIVER CAN 3V3

CAN_TX/RX: interfaccia ESP: CAN_H/L: lato rete CAN.



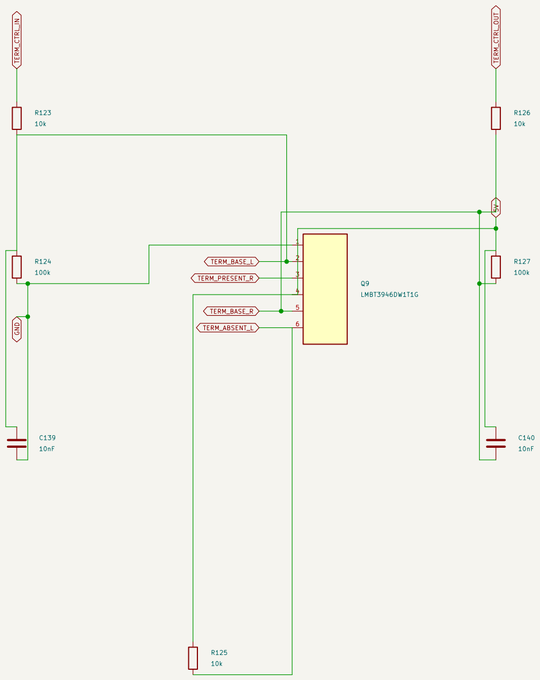
INTERRUTTORE E RESISTENZA DI TERMINAZIONE

TERM_DRIVE / TERM_LED: pilotaggio foto-MOS.
TERM_R: collegamento tra120ohm e interruttore verso CAN_L.
Quando abilitata, la resistenza120ohm e fra CAN_H e CAN_L.



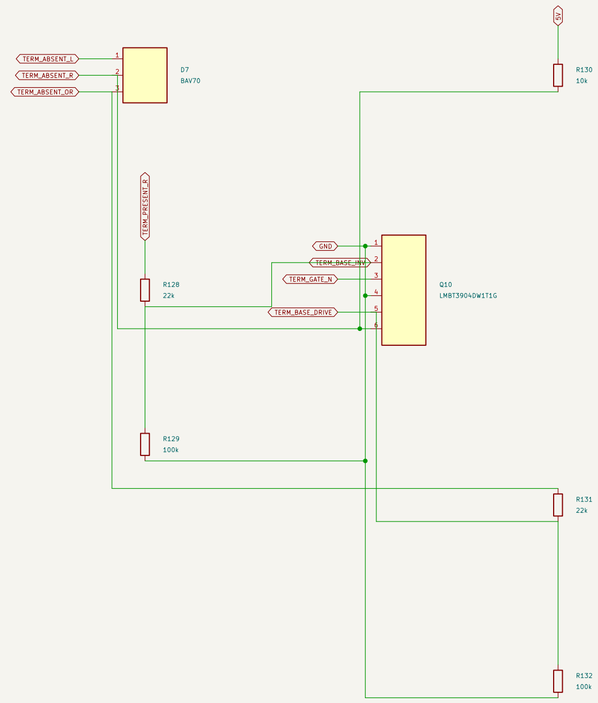
RILEVAMENTO VICINO SINISTRO / DESTRO

TERM_BASE_L/R: nodi di filtro.
TERM_ABSENT_L: assenza a sinistra; TERM_PRESENT_R: presenza a destra.

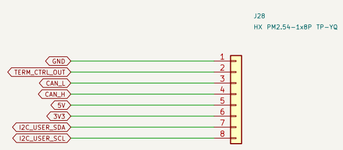


COMBINAZIONE ASSENZE E COMANDO

TERM_ABSENT_R e TERM_ABSENT_OR: estremità rilevate.
TERM_GATE_N: comando attivo basso del PMOS di terminazione.

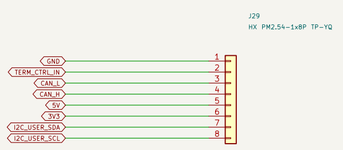


ESPANSIONE BUS J28 / MAGNETICO J9



Pin 1..5 condividono i fori e le reti del magnetico.
6:3V3 / 7:SDA / 8:SCL. I2C solo locale a3.3V.
DET mantiene la funzione originale di terminazione CAN:
non usarlo come GPIO e non forzarlo sulla espansione.
Non alimentare contemporaneamente il bus da sorgenti
non isolate. Nessuna protezione aggiunta su questi pin.

ESPANSIONE BUS J29 / MAGNETICO J2



Pin 1..5 condividono i fori e le reti del magnetico.
6:3V3 / 7:SDA / 8:SCL. I2C solo locale a3.3V.
DET mantiene la funzione originale di terminazione CAN:
non usarlo come GPIO e non forzarlo sulla espansione.
Non alimentare contemporaneamente il bus da sorgenti
non isolate. Nessuna protezione aggiunta su questi pin.

Sheet: /03 – CAN e terminazione automatica/
File: can.kicad_sch

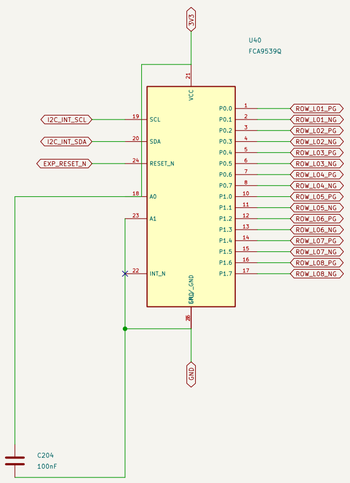
Title: 03 – CAN e terminazione automatica

Size: User	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 3/9

CONTROLLO48SEGNALI / RESET SICURO / I2C

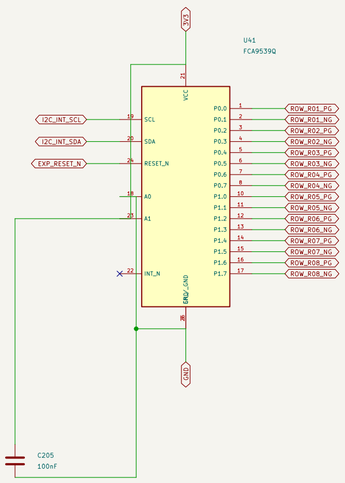
U40 | FCA9539Q | 0x75

PG/NG banco A: comandi di alimentazione.
SCL/SDA: I2C: A0/A1: indirizzo: INT_N non usato.



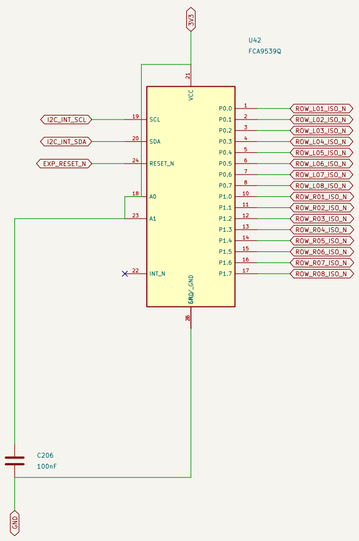
U41 | FCA9539Q | 0x76

PG/NG banco B: comandi di alimentazione.
SCL/SDA: I2C: A0/A1: indirizzo: INT_N non usato.



U42 | FCA9539Q | 0x77

ISO_N A/B: isolamento GPIO, attivo basso.
SCL/SDA: I2C: A0/A1: indirizzo: INT_N non usato.



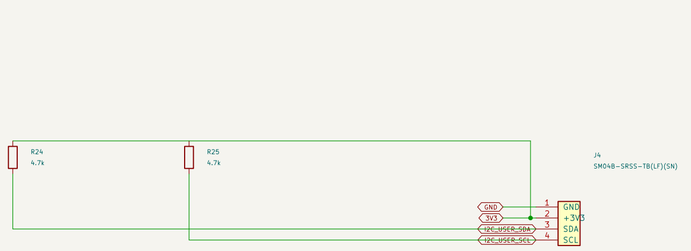
INTERBLOCCO RESET / UNA SOLA PORTA

EXP_RESET_N = ESP_EN AND NOT GPIO0. INO aGND.
R203 mantiene reset basso se non pilotato: C200 bypass100nF.

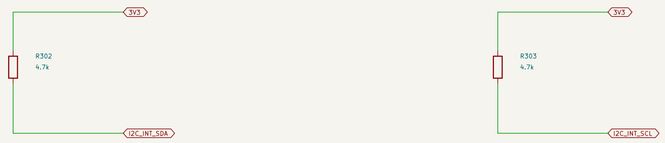


BUS UTENTE GPIO11/12 / QWIIC / PULL-UP

I2C_USER indipendente dall'expander: SDA GPIO11 / SCL GPIO12. Pull-up4.7k a3V3.
3V3/GND: alimentazione del moduloQwiic.



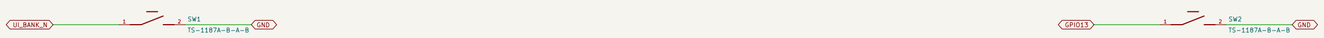
BUS INTERNO DEDICATO / GPIO4 SDA / GPIO5 SCL



R302/R303: pull-up4.7k del solo bus interno.
R24/R25: pull-up4.7k del bus utente, nel blocco Qwiic.
Nessun collegamento SDA/SCL tra i due bus.
Entrambi usano livelli3.3V: firmware: due controller distinti.

INTERFACCIA LOCALE | DUE PULSANTI XKB

SW1 SELECT: GPIO21 / UI_BANK_N. SW2 OK: GPIO13. Inputs with internal pull-ups: debounce25ms.
SELECT selects action: OK opens slots1..4. SELECT chooses slot. OK confirms.
Hold OK2s for DELETE. SELECT+OK stops immediately: suppress singles until release.
GPIO13 is shared with exported connectors and reserved for OK. No extra input IC.



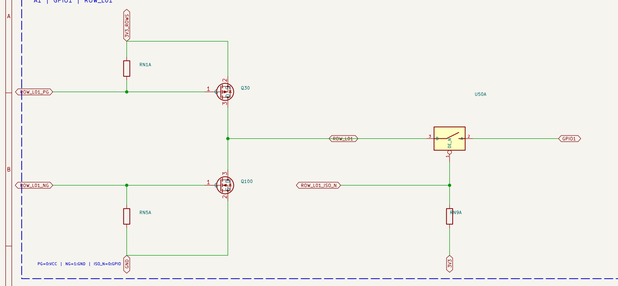
Sheet: /04 – Controllo righe, reset sicuro e Qwiic/
File: control.kicad_sch

Title: 04 – Controllo righe, reset sicuro e Qwiic

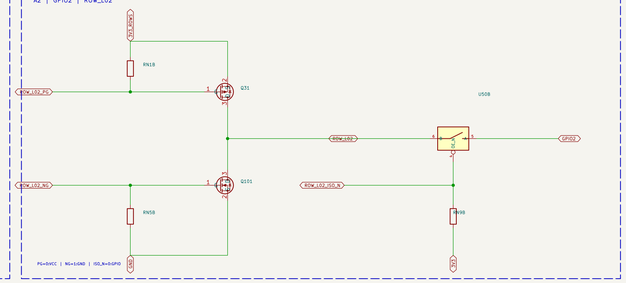
Size: User | Date: 2026-11-08 | **Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3 | Id: 4/9

16 RIGHE PROGRAMMABILI | VCC3V3 / GND / GPIO / HI-Z

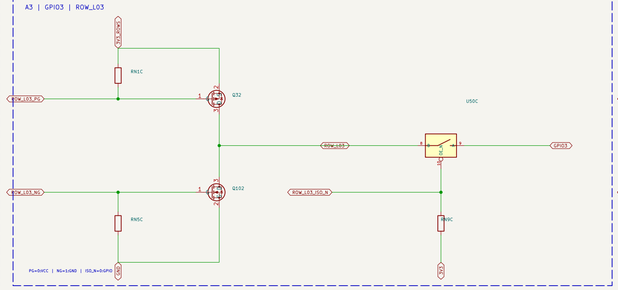
A1 | GPIO1 | ROW_L01



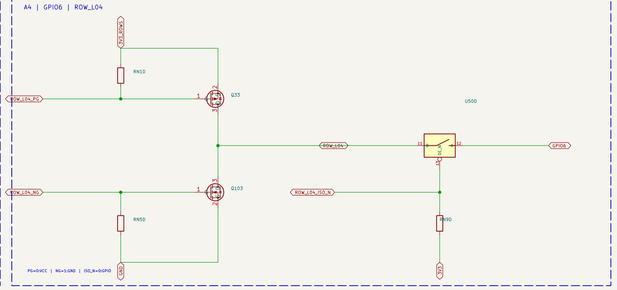
A2 | GPIO2 | ROW_L02



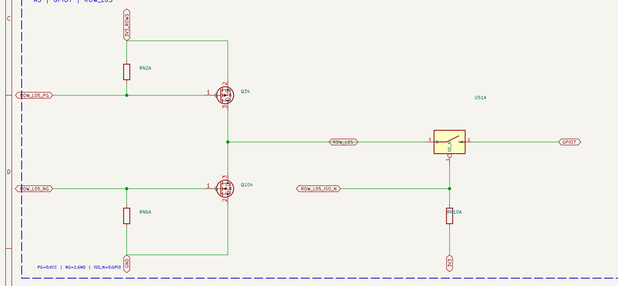
A3 | GPIO3 | ROW_L03



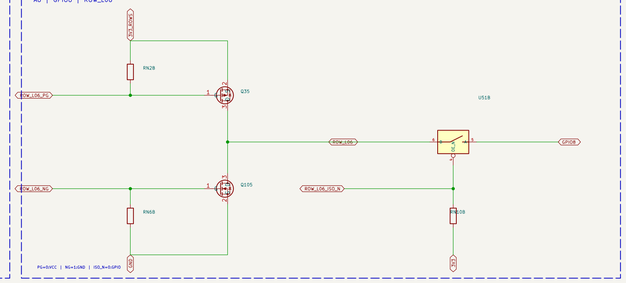
A4 | GPIO6 | ROW_L04



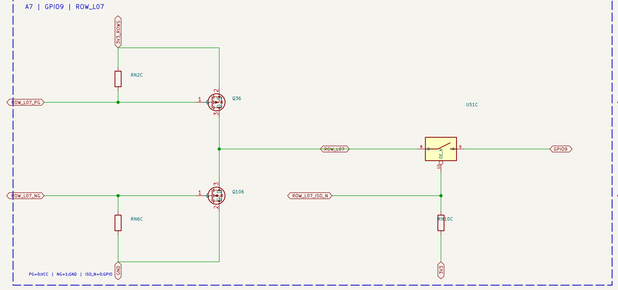
A5 | GPIO7 | ROW_L05



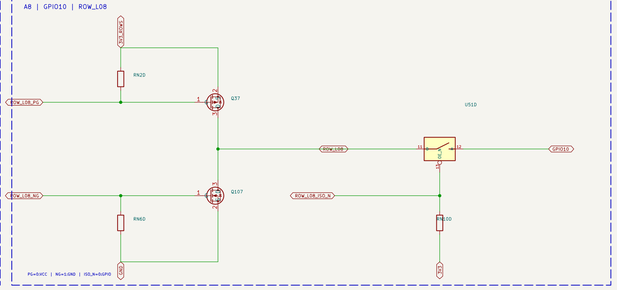
A6 | GPIO8 | ROW_L06



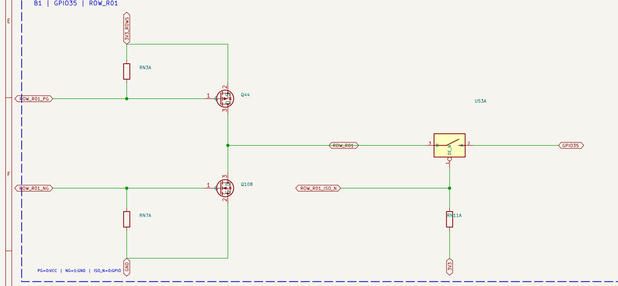
A7 | GPIO9 | ROW_L07



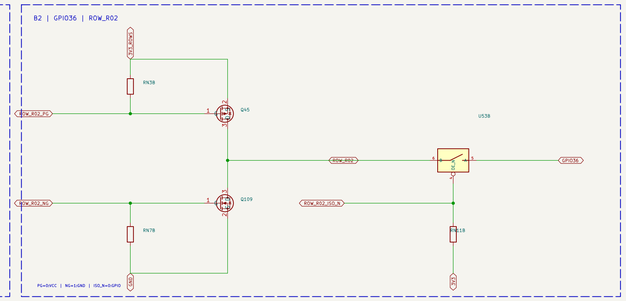
A8 | GPIO10 | ROW_L08



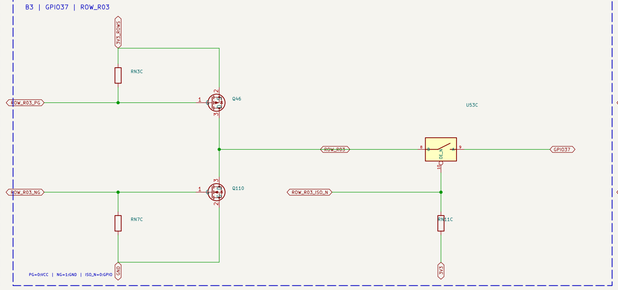
B1 | GPIO35 | ROW_R01



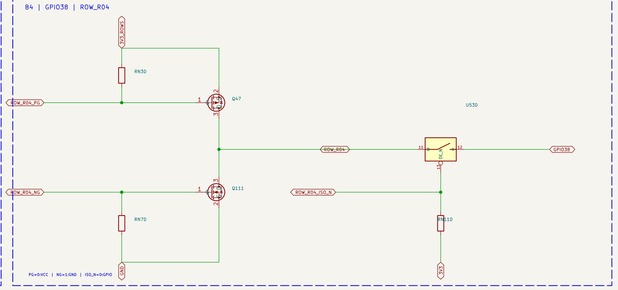
B2 | GPIO36 | ROW_R02



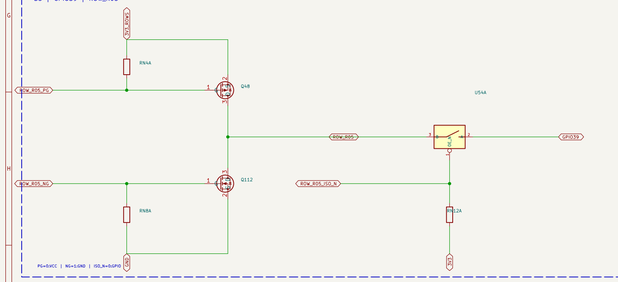
B3 | GPIO37 | ROW_R03



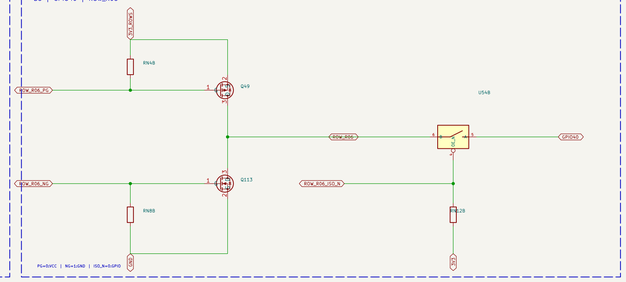
B4 | GPIO38 | ROW_R04



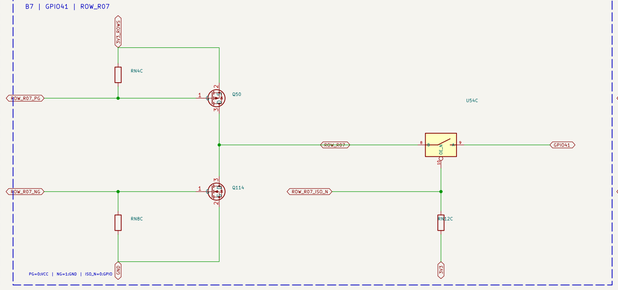
B5 | GPIO39 | ROW_R05



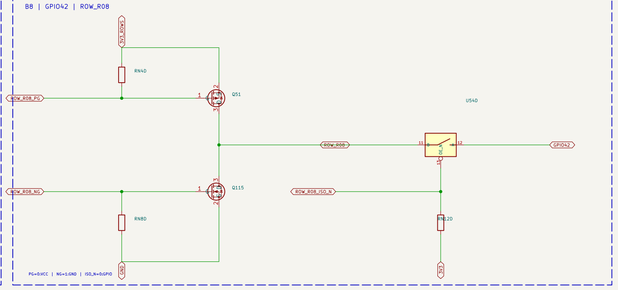
B6 | GPIO40 | ROW_R06



B7 | GPIO41 | ROW_R07

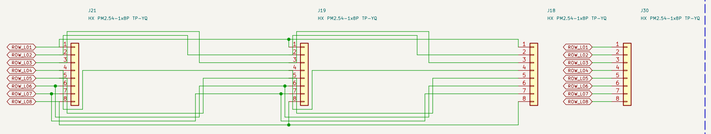


B8 | GPIO42 | ROW_R08



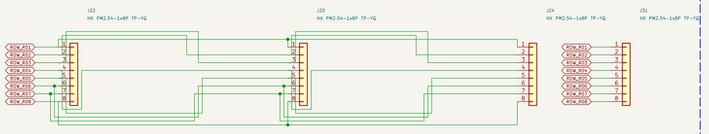
SOCKET BANCO A | 4CONTATTI COMUNI PER RIGA

ROW_L01..08=A1..A8. ROW_R01..08=B1..B8. Passo righe2.54mm; colonne4mm; separazione centrale7.62mm.



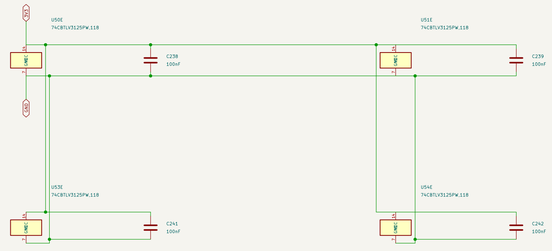
SOCKET BANCO B | 4CONTATTI COMUNI PER RIGA

ROW_L01..08=A1..A8. ROW_R01..08=B1..B8. Passo righe2.54mm; colonne4mm; separazione centrale7.62mm.



ALIMENTAZIONE DEI4ISOLATORI

Ogni integrato bancasali. UnitaE=alimentazione: bypass100nF.



REGOLA: PG=1, NG=0: ISO_N=1 -> Hi-Z. Per ADC: disattivare entrambi i MOSFET, utilizzare solo ISO_N e ingresso analogico.
Mai PG=1 e NG=1 (corto tra +3.3_ROW e GND). Aggiungere interblocco hardware o firmware.
Q30..Q51 = AO3401A (P); Q100..Q115 = AO3400A (N). Switch bidirezionali TS5DP. Invertali.
Obiettivo 500 mA su singola riga in VCC/GND: budget condiviso, piste da abbigliare e prova termica necessaria. GPIO/ADC non sono uscite da 500 mA.

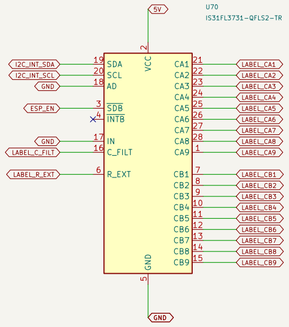
Sheet: /05 – Sedici righe programmabili/
File: ports16.kicad_sch

Title: 05 – Sedici righe programmabili

Size: A1	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 5/9

DRIVER5V | I2C INTERNO3,3V |48LED

is31fl3731_0x74: SDB=ESP_EN3,3V. Nessun nuovo GPIO. R_EXT47kx1.38mAmedi/LED aPWM255.
CA16rossi, CB:16blu+16verdi. Avvio in software shutdown; azzerare enable/PWMprima di accendere.











A1 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A2 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A3 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A4 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A5 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A6 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A7 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A8 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B1 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B2 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B3 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B4 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B5 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B6 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B7 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B8 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

Sheet: /06 – Indicatori di configurazione retroilluminati/
File: rgb-ring.kicad_sch

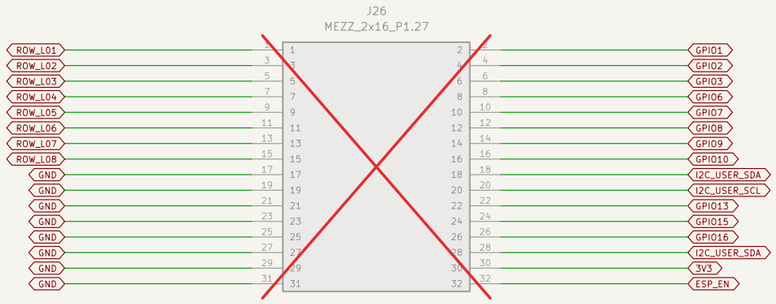
Title: 06 – Indicatori tre colori,5V,testi dritti

Size: A2	Date: 2026-11-08	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 6/9

J26 / BANCO A: FORI E GPIO SEPARATI

1..16: ROW = fori; GPIO = lato ESP, prima dello switch.

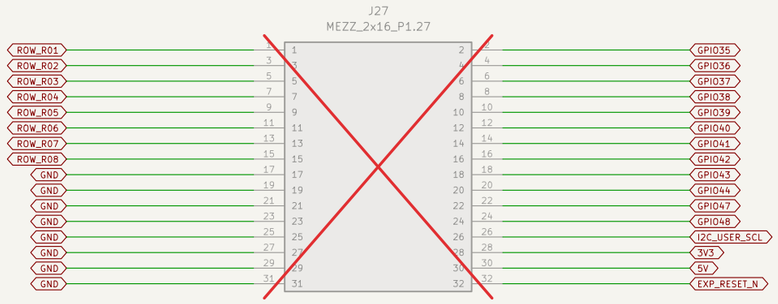
17..32: espansione, rail, masse e reset.



J27 / BANCO B: FORI E GPIO SEPARATI

1..16: ROW = fori; GPIO = lato ESP, prima dello switch.

17..32: espansione, rail, masse e reset.



BACKBONE PASSIVA: aggiunti solo J26/J27: nessun ADC/DAC, selettore, shunt o protezione extra.
Mezzanina assente: prese aperte. Presente: ingressi/uscite analogiche da progettare ad alta impedenza e sicuri anche senza alimentazione.
GPIO e ROW non vanno uniti sulla mezzanina: bypasserebbero gli switch e la separazione dal3V3/GND delle righe.
GPIO3.3V, non tolleranti5V. Righe limitate al dominio della backbone. I2C_USER condiviso con Qwiic/Hanxia, separato dagli expander; evitare conflitti di indirizzo.
ESP_EN: reset ESP, non abilitazione dedicata. EXP_RESET_N: uscitaU60, solo lettura; non comandarla dalla mezzanina.
Sincronizzazione e abilitazione: assegnabili nel firmware ai7GPIO liberi esposti anche suJ25; GPIO11/12 riservati al bus utente, non a pin aggiuntivi dedicati.
La mezzanina di misura deve avere protezioni, disconnessione a riposo e sequenze sicure proprie; nessuna protezione nuova sulla base.
La corrente si misura instradando il circuito fra due righe distinte attraverso lo shunt della mezzanina; nessuno shunt sulla backbone.

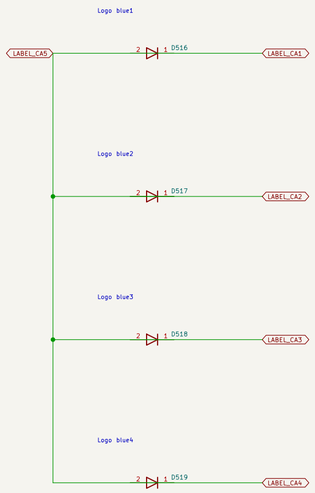
Sheet: /07 – Mezzanina passiva/
File: mezzanine.kicad_sch

Title: 07 – Mezzanina passiva / tutti i nodi e GPIO

Size: A3	Date: 2026-11-08	**Rev: MEZZ-PASSIVE**
KiCad E.D.A. 10.0.3		Id: 7/9

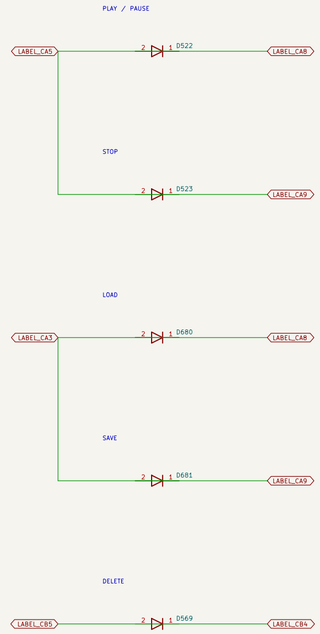
BLUE LOGO

Four retained blue emitters; same driver and channels.



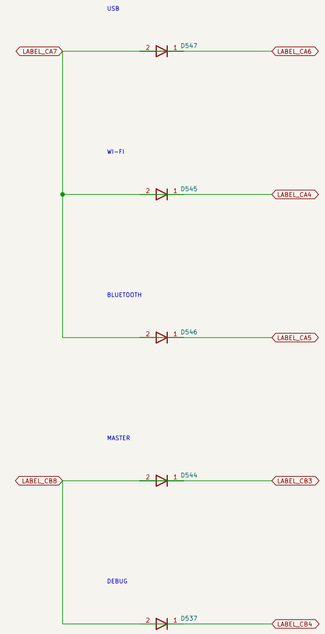
SELECT / OK ACTION MENU

PLAY/PAUSE, STOP, LOAD, SAVE, DELETE. DEBUG is status only.



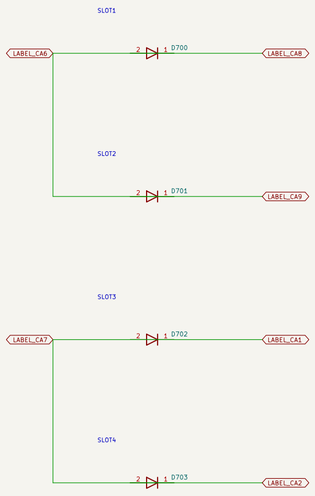
CONNECTION AND ROLE STATUS

Indicators are not extra touch inputs.



FOUR CONFIGURATION SLOTS

D700..D703: green, existing unused IS31FL3731 channels.
1..4 selected by SELECT, confirmed with OK.



CAN TERMINATION INDICATOR

Existing direct drive, independent of the Charlieplex driver.

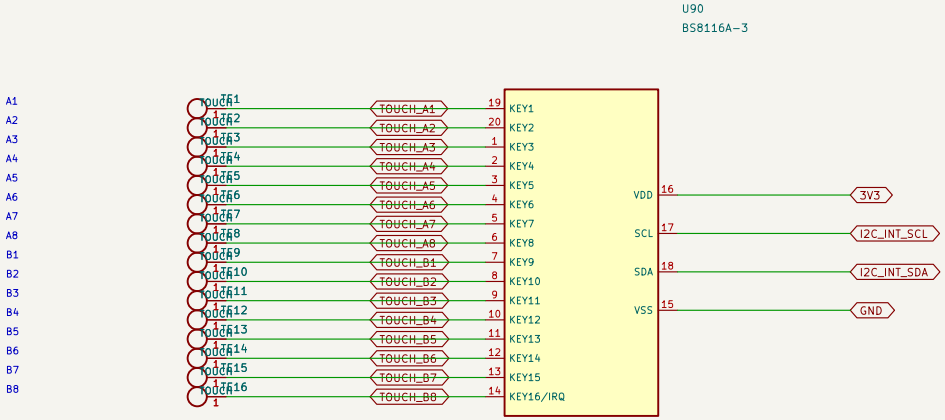


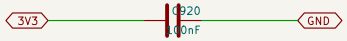
Hold OK2s deletes only the selected saved slot. It does not erase the working draft.
LOAD validates first, disarms outputs, loads the draft and requires PLAY to apply it.
Navigating a menu never changes outputs. Both buttons always request STOP.

Sheet: /08 – Selezione, comandi e logo/	
File: ui-indicators.kicad_sch	
Title: 08 – UI status and four configuration slots	
Size: A2	Date: 2026-11-09
KiCad E.D.A. 10.0.3	**Rev: 16R-SELECTOK4**
	Id: 8/9

ONE CONTROLLER / SIXTEEN ROW KEYS

U90 BS8116A-3, address0x50 on internal I2C. No multiplexer.
Key16 defaults to IRQ: clear K16_TH bit6 (0xC4), use polling. Follow full configuration-write protocol.





Global actions use SW1/SW2, not these touch channels.
Touch discrimination, full routing and final hardware qualification remain open.

Sheet: /09 – Row touch,16keys/
File: touch16.kicad_sch

Title: 09 – Sixteen row touch keys

Size: A4	Date: 2026-11-09	**Rev: 16R-SELECTOK4**
KiCad E.D.A. 10.0.3		Id: 9/9